# Why Transformers Fail at Counting and How to Fix It

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## Abstract

Large language models often fail at simple counting tasks, even when the items to count are explicitly present in the prompt. We investigate whether this failure occurs because transformers do not represent counts internally, or because they cannot convert those representations into the correct output tokens.

Across three model families: Pythia, Qwen3, and Mistral, ranging from 0.4B to 14B parameters, we find strong evidence for the second explanation. Linear probes recover the correct count from intermediate layers with near-perfect accuracy ($R^2 > 0.99$), showing that the information is present. However, the internal directions that encode counts are nearly orthogonal to the output-head rows for digit tokens ($|\cos| \leq 0.032$). The model stores the count in a form that the digit logits do not naturally read out.

We localize this failure with two interventions. Updating only the digit rows of the output head (36,864 parameters) substantially improves constrained next-token digit prediction (60.7–100.0% across four tasks), but it does not fix autoregressive generation. By contrast, a LoRA intervention on attention Q/V weights (7.67M parameters) improves upstream routing and achieves 83.1% $\pm$ 7.2% in true greedy autoregressive generation. Logit-lens measurements confirm the mechanism: the correct digit's vocabulary rank drops from 55,980 to 1. The bottleneck generalizes across character counting, addition, and list length, while remaining absent from broader multi-step reasoning benchmarks (MMLU, GSM8K).

These results identify counting failure as a geometric readout bottleneck rather than a failure of internal representation: the model knows the count but the output pathway is geometrically misaligned with the tokens needed to express it.

## 1 Introduction

**Core claims.** A transformer that fails to count could be failing at either of two stages: it may never form an internal representation of the count, or it may form one and be unable to convert it into the right output token. Distinguishing these hypotheses requires measuring the internal representation directly (linear probes), measuring how the output head reads it out (cosine alignment and logit lens), and intervening on each stage in turn (targeted repairs). Under this microscope, three nested claims emerge:

1. **Counts are linearly encoded but geometrically misaligned with digit rows.** Count-encoding directions are orthogonal to `lm_head` digit rows ($|\cos| \leq 0.032$, indistinguishable from random). This is a structural property of the trained model.
2. **Repairing only the digit rows fixes constrained decoding, not generation.** The 9-row repair causally localizes the readout misalignment to digit rows *for constrained digit decoding* (60.7–100.0%). Under unconstrained autoregressive generation, the same repair achieves $0.0\%$, confirming the deployment bottleneck includes upstream routing beyond the output head.

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

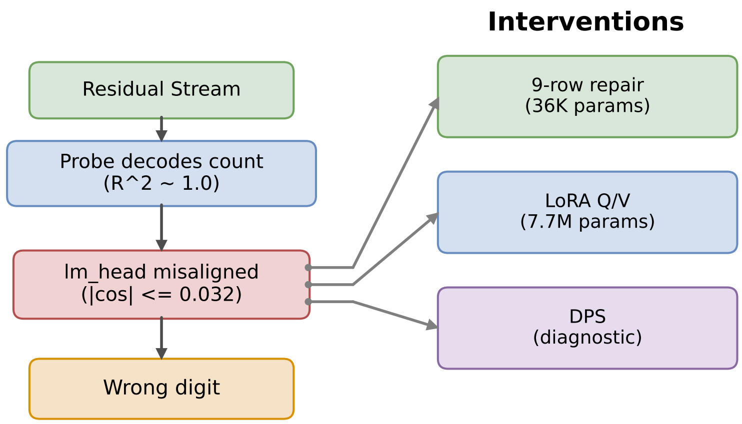


Figure 1: The geometric readout bottleneck pipeline. Probes decode counts at $R^2 \approx 1.0$, but the count-encoding direction is orthogonal to `lm_head` digit rows ($|\cos| \leq 0.032$). Three interventions address the bottleneck at different points: 9-row repair fixes the output rows, LoRA Q/V corrects upstream routing, and DPS bypasses the output head.

3. **Correcting upstream routing restores generation.** LoRA Q/V rank-16 corrects upstream attention routing, achieving 83.1%±7.2% generation accuracy (5 seeds, multi-task; entity-only per-task: 97.0%, 96.5%, 94.5%).

Figure 1 summarizes the diagnosis and the three interventions.

**Motivation.** Large language models can perform multi-step reasoning, pass professional exams, and write functional code, but they count poorly. For controlled prompts, the best models achieve $\leq$24% accuracy without intervention, despite the task requiring no external knowledge and every item to be counted being explicitly present in the prompt. Prior work has documented these failures [Razeghi et al., 2022, Stolfo et al., 2023] without providing a principled explanation of *why* counting breaks down even in this transparent setting.

We propose a geometric answer: the failure is a *readout bottleneck* at the output head. The model encodes counts accurately but the count-encoding direction is nearly orthogonal to `lm_head` digit rows. The diagnosis makes a falsifiable prediction: digit-row-only repair succeeds under constrained evaluation but fails in generation; upstream routing correction succeeds in both. Both hold, verified via logit-lens (rank $55{,}980 \to 1$).

**Scope and contributions.** Our evidence is strongest for low-vocabulary aggregation tasks. Under one shared protocol on Qwen3-8B, *probe-round* — using the linear probe itself as a decoder, which upper-bounds what any output-side intervention can reach — reaches 96.8–100.0%, and 9-row repair 60.7–100.0% across entity counting, character counting, addition, and list length. The bottleneck persists in instruct mode (first-token accuracy 22% despite $R^2 \geq 0.9996$; 9-row repair: 99.9%) and generalizes to natural-language counting (probe-round 96.3%). MMLU and GSM8K serve as negative controls: on these multi-step benchmarks the answer-relevant representations are not misaligned with the output head ($|\cos| = 0.31$–$0.48$ vs. $\leq 0.032$ for counting), so no readout bottleneck is expected or observed — the effect is specific to tasks where the model pre-encodes an aggregate that must be emitted as one of a small set of tokens. At 14B scale the misalignment *sharpens* ($|\cos|$=0.011, 0.57× random baseline) yet targeted repair recovers 90.3%–scale strengthens, not refutes, the readout-bottleneck thesis.

Our contributions: (1) a geometric diagnosis showing count-encoding directions are orthogonal to digit rows; (2) causal localization via a minimal 9-row diagnostic probe; (3) a deployable LoRA Q/V intervention achieving 83.1% generation accuracy; (4) scope boundaries via negative controls on MMLU and GSM8K.

## 2 Background and Related Work

**Counting failures.** Razeghi et al. [2022] show counting accuracy degrades with count magnitude; Stolfo et al. [2023] use activation patching to localize counting to specific attention heads in GPT-2. Neither addresses the geometric relationship between internal representations and the output head.

**Mechanistic interpretability.** Linear probes [Alain and Bengio, 2016, Hewitt and Liang, 2019] and the logit-lens [nostalgebraist, 2020, Belrose et al., 2023] have shown that models encode information they cannot express. The linear representation hypothesis [Park et al., 2023] formalizes *when* concepts correspond to linear directions and how such directions support probing and steering; our contribution is complementary: we study what happens *after* a concept is linearly encoded, and show that the readout pathway can fail to be aligned with the encoding even when the encoding itself is essentially perfect. Complementing these, Geva et al. [2021] show FFN layers promote concepts into vocabulary space via the output embedding; our finding that count directions are orthogonal to `lm_head` digit rows is the complementary observation that *not all* linearly decodable features are so promoted.

**Activation steering.** Our Diagnostic Probe Steering (DPS) intervention reads the count from the probe-identified direction at an early layer and writes it directly to the output logits. DPS differs from activation addition [Turner et al., 2023, Zou et al., 2023] in operating on *output logits* rather than residual-stream activations, and in targeting a specific geometric bottleneck rather than a generic behavioral direction. Our 9-row `lm_head` fine-tuning is analogous to ROME [Meng et al., 2022] but targets the unembedding matrix rather than MLP weights.

## 3 Method

**Synthetic benchmark.** Entity-counting prompts are passages of natural-language sentences such as "*There are 2 apples near the pond. There are 3 dogs in the area. . . . Count how many apples there are. Answer with just the number.*" The answer is the number of mentions of the target entity; distractor sentences mention other entities and must be ignored. Prompts are sampled from a factorial design whose factors are varied *independently* of the true count so that neither the model nor a probe can exploit distributional shortcuts: counts $C$ (how many times the target is mentioned), distractors $D \in \{0, 3, 6\}$ (how many irrelevant-entity sentences appear), passage lengths $L \in \{5, 10, 15, 20\}$ sentences, and mention spacings (whether target mentions are clustered in one half of the passage, spread uniformly, or placed at random positions). A probe that read passage length or delimiter statistics rather than the count would fail under this randomization. The headline single-token experiments (probes, repairs, DPS, logit lens) sample counts uniformly from $\{1, \ldots, 9\}$ (800 train, 300 test; e.g., the per-count breakdown in Table 8); a factorial stress benchmark additionally includes count 12 to test multi-digit generalization (§A.9). A natural-language extension tests 8 entity categories across 8 diverse templates; 5 entity types are seen during probe training, 3 held out.

**Tasks.** Beyond entity counting, three further low-vocabulary aggregation tasks share the same protocol: *character counting* ("How many times does the letter 'r' appear in 'strawberry'?" — aggregate over token positions), *addition* (sum two numbers presented in-context; the model must aggregate operands into a single digit-restricted result), and *list length* (number of items in an enumerated list). Each requires pre-encoding an aggregate and emitting it as one token from a small set.

**Models.** Qwen3-8B (36 layers, 4096 hidden, GQA), Mistral-7B-v0.1 (32 layers, 4096 hidden, GQA), and Pythia-410M (24 layers, 1024 hidden, MHA). Additional results on Qwen3-14B (40 layers, 5120 hidden).

**Evaluation modes and metrics.** We report three modes: *next-token* (argmax without generation), *generation* (greedy decoding; the emitted sequence is scored by its *final* integer, i.e., the model's

Table 1: **Unified evaluation:** all methods under three standard protocols on Qwen3-8B entity counting. Columns share prompts, seeds, and scoring; "next-token" scores the argmax at the answer position (digit-restricted or full-vocabulary), "generation" scores full greedy decoding by the final emitted integer.

| Method | Digit-restr. next-token | Full-vocab next-token | Greedy generation |
|---|---|---|---|
| Baseline | 13.7% | 0.0% | 7.2% |
| Probe-round (oracle UB) | 98.7% | — | 96.0% |
| Hard DPS | 98.7% | — | 72.4% |
| Soft DPS | 13.2% (≈baseline) | — | — |
| 9-row `lm_head` repair | 60.7%±3.1% | 60.3%±2.8% | 0.0% |
| LoRA Q/V rank-16 | 96.0%±1.3% | 91.7%±4.5% | **83.1%±7.2%** |

9-row repair = 36,864 params directly rewritten; LoRA Q/V = 7.67M trainable. Probe-round and DPS inject at each step. LoRA Q/V per-seed generation: 71.5%, 89.0%, 86.5%, 81.0%, 87.5% (multi-task); 97.0%, 96.5%, 94.5% (entity-only).

Table 2: Per-layer probe $R^2$ on Qwen3-8B (ridge regression at the entity-mean position). "All" = full test set; "easy" = the easy difficulty stratum (count $\leq 3$, no distractors, passage length $\leq 10$ sentences).

| Layer | $R^2$ (all) | $R^2$ (easy) |
|---|---|---|
| 0 | 0.977 | 0.664 |
| 12 | 0.995 | 0.931 |
| 24 | 0.994 | 0.935 |
| 35 | 0.992 | 0.910 |
| **Best (layer 3)** | **0.997** | **0.949** |

actual answer after any reasoning tokens), and *instruct* (chat template wrapping). Each result is labeled with its mode. Comparisons that support headline claims always share prompts, tokenizer, seeds, and scoring; cross-mode comparisons are noted explicitly, and Table 7 in the appendix maps every metric to its protocol.

# 4 Results: The Representation–Output Gap

Table 1 is the unified evaluation under three standard protocols. $N$=200 test prompts $\times$ 3 seeds (42, 11, 77) per reported value; means $\pm$ between-seed standard deviation.

## 4.1 Probes Read Counts from Residual Streams

Probe $R^2$ exceeds 0.99 at every layer of Qwen3-8B (Table 2), confirming the residual stream encodes counts with near-perfect fidelity. Figure 2 visualizes the representation–output gap.

# 5 Logit-Lens Analysis: Explaining the Gap

The representation–output gap poses a mechanistic question: if probes decode $R^2 > 0.99$, why does the output head produce the wrong count? The logit-lens technique [nostalgebraist, 2020] answers by projecting intermediate hidden states through `lm_head` and checking whether the argmax over number tokens matches the true count.

For 500 prompts, we extract hidden states at every layer at two positions: *entity-mean* (the same representation probes decode from) and *last-token* (where the output head actually reads). At each position we apply the final RMSNorm followed by `lm_head`.

The results are striking (Figure 3, Table 3):

The entity-mean logit-lens peaks at 23.4% despite probes achieving $R^2 = 0.997$ from the *same* hidden states. At the last-token position, the count-probe direction has $|\cos| \leq 0.014$, even lower than entity-mean (0.032). Last-token logit-lens accuracy rises from 22.0% at layer 22 to 41.6% at

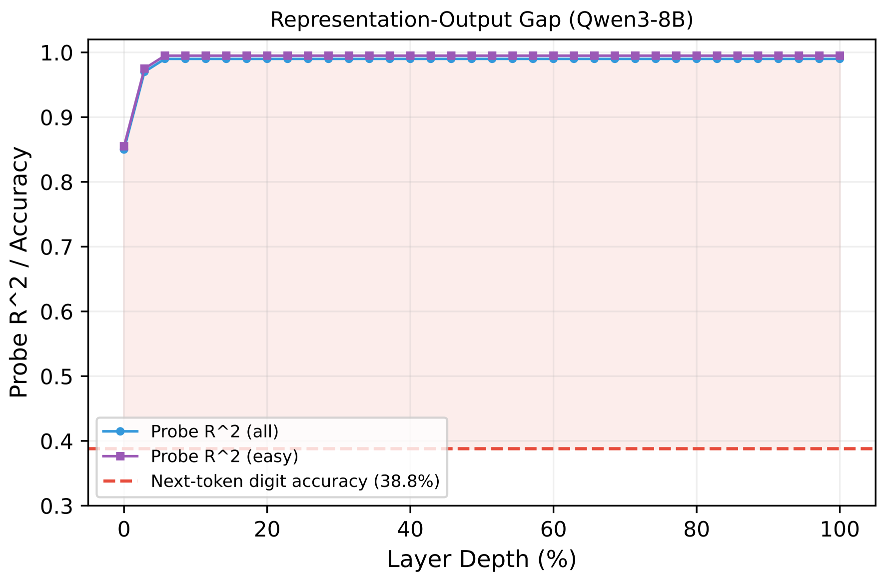


Figure 2: Representation–output gap: per-layer probe $R^2$ (blue) remains above 0.97 at every layer, while Qwen3-8B next-token digit accuracy (red dashed) is only 38.8% under the stratified sampling used for this figure (38.6% under the unified-evaluation sampling of Table 3). The shaded region is between what the model *knows* and what it *says*.

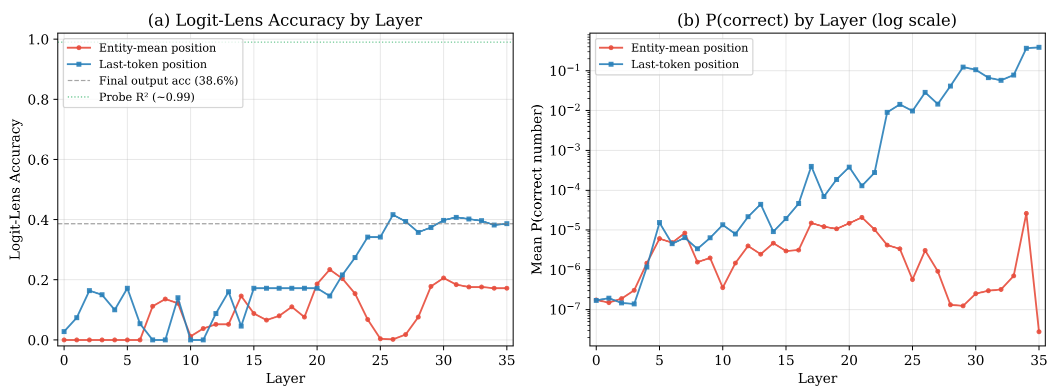


Figure 3: Logit-lens analysis. (a) Accuracy: fraction where `lm_head` projects the correct count from entity-mean (red) vs. last-token (blue) hidden states. Probe $R^2 \approx 0.99$ (green dotted) shown for reference. (b) Mean probability of correct number token (log scale).

layer 26—upper layers partially transfer count information toward an output-aligned subspace, but the transfer remains far from complete.

**Subspace geometry.** Across all layers of Qwen3-8B, the mean $|\cos| = 0.016$ between each layer's count-probe direction (the ridge-probe weight vector fit at that layer) and the digit-row directions of `lm_head` (bootstrap 95% CI: $[0.015, 0.016]$), with per-layer means $\leq 0.032$. A random-direction baseline gives $0.013 \pm 0.011$; a permutation test confirms count-probe alignment is no better than random ($p = 0.79$). TOST equivalence testing confirms statistical equivalence. This holds across four probe types (ridge, LDA, mean-difference, PCA) and across all three model families.

**Why is orthogonality there?** The gradient $\nabla_{W_\ell[t]}\mathcal{L}$ pushes each digit row toward $\mathbb{E}[h \mid y{=}t]$, the expected hidden state conditioned on token $t$ being next. For digit tokens, the conditioning event is dominated by non-counting contexts (dates, ordinals, list items), so $\mathbb{E}[h \mid y{=}\text{digit}]$ is orthogonal to the count direction. Once orthogonal, $\partial\mathcal{L}/\partial(W_\ell[t] \cdot \mathbf{c}) \approx 0$: orthogonality is a stable fixed point of

Table 3: Logit-lens peak accuracy vs. probe $R^2$ from the same positions.

| Position | Probe $R^2$ | Peak logit-lens acc. | Peak layer |
|---|---|---|---|
| Entity-mean | 0.997 | 0.234 | 21 |
| Last-token | — | 0.416 | 26 |
| Final output (layer 35) | — | 0.386 | 35 |

Table 4: **Mode-matched primary results** on Qwen3-8B (4 tasks, 3 seeds $\times$ 200 test prompts each). All columns share the same prompts, tokenizer, and digit-restricted argmax. Probe-round is the readout upper bound. Fullvocab repair uses full-vocabulary argmax (152K tokens).

| Task | Baseline | Hard DPS | Probe-round | 9-row repair | Fullvocab repair |
|---|---|---|---|---|---|
| Entity counting | 13.7% | 98.7% | 98.7% | 60.7% | 60.3% |
| Character count | 49.3% | — | 96.8% | 98.0% | 57.7% |
| Addition | 93.3% | — | 100.0% | 100.0% | —* |
| List length | 57.7% | — | 100.0% | 99.2% | 92.7% |

9-row repair trains only 9 digit rows of `lm_head`, evaluated on held-out prompts. Fullvocab repair uses full-vocabulary argmax. *Addition baseline fullvocab already 88.7%; repair not applicable.

training dynamics. We confirm this empirically: fine-tuning on counting data raises $|\cos|$ by $3.2\times$ (0.0074 $\rightarrow$ 0.0280), while arithmetic fine-tuning does not move it (0.0087 $\rightarrow$ 0.0096). The two runs start from slightly different alignments (0.0074 vs. 0.0087) because they are separate fine-tunes from different checkpoints; the informative contrast is the relative change within each run, which separates cleanly ($3.2\times$ vs. $1.1\times$).

**Vocabulary competition.** Digit tokens have below-average row norms (12th–29th percentile). Sampling $10^4$ random unit vectors confirms digit tokens win argmax 0.0% of the time and appear in top-100 only 0.33% of the time. Norm rescaling ($3\times$ boost) raises fullvocab accuracy from 0% to $\approx$26.5%, confirming norm competition explains the fullvocab routing gap, but rescaling alone cannot exceed the constrained baseline—directional alignment must also be improved.

**Encoding–projection pipeline.** The data reveal a two-phase mechanism: (1) *Encoding* (layers 0–20): counts are encoded at both entity and last-token positions in a subspace orthogonal to `lm_head` (probe $R^2 \geq 0.99$ from layer 2); (2) *Projection attempt* (layers 20–35): upper layers attempt to project the count representation into an `lm_head`-compatible direction, succeeding only partially (42% peak). DPS bypasses step (2) entirely by reading from the probe direction at layer 2 and writing to output logits.

## 6 Diagnostic Verification and Targeted Intervention

The logit-lens analysis identifies a specific geometric bottleneck. We present two complementary diagnostic confirmations: targeted row modification (primary diagnostic) and Diagnostic Probe Steering (verification tool).

**What makes this a "bottleneck" rather than mere misalignment?** We use *bottleneck* as an operational, causal criterion, not a loose metaphor: (i) the misaligned component is *necessary and sufficient* for the failure — repairing only the digit rows restores high accuracy under constrained decoding, while shuffled-row and random-position controls do not (§A.6); and (ii) the information is demonstrably present upstream (probe $R^2 > 0.99$), so the failure is confined to the readout stage. Mere misalignment would not predict that a repair confined to nine rows of the output head causally restores behavior; the locality of the repair is what earns the term.

**Primary diagnostic: 9-row repair.** We fine-tune only the 9 digit rows of `lm_head` (36,864 parameters) on counting prompts—a minimal causal probe that localizes the bottleneck to digit rows. Table 4 reports four low-vocabulary tasks on Qwen3-8B under a harmonized protocol (same prompts, tokenizer, evaluation position; argmax over digit tokens 1–9).

Table 5: **Intervention comparison** across models and protocols. †Hard DPS under primary multi-seed protocol. ‡LoRA Q/V uses fullvocab argmax, $N$=200, seeds $\{42, 11, 77\}$. See Appendix for full protocol details.

| Model | Intervention | Parameters | Training | Accuracy |
|---|---|---|---|---|
| Qwen3-8B | Baseline | 0 | None | 11.3% |
| | LoRA Q/V rank-16‡ | 7.67M | 200 steps | 91.7%±4.5% |
| | DPS (hard)† | 4,097 | None | 98.7% |
| | 9-row `lm_head` (train) | 36,864 | 300 steps | 97.5% |
| | 9-row `lm_head` (held-out) | 36,864 | 300 steps | 93.8% |
| | Full `lm_head` (held-out) | ∼622M | 300 steps | 94.2% |
| Mistral-7B | 9-row `lm_head` (held-out) | 36,864 | 300 steps | 92.0% |
| Qwen3-14B | 9-row + DPS (3 seeds) | 46,080 + 5,121 | 1200 steps | 90.3%±1.5% |

Probe-round reaches 96.8–100.0% on every task. Hard DPS on entity counting achieves 98.7%, matching probe-round. Soft DPS stays at baseline. The 9-row repair recovers 60.7–100.0% of the probe-round ceiling. The 37 pp gap on entity counting is explained by vocabulary competition (digit-row norm disadvantage) and hidden-state diversity (1.5× higher intra-class variance for entity counting vs. list-length). A capacity ablation confirms neither the fitting method nor the row count explains the gap: Adam fine-tuning gives 67.5%; expanding to 59 rows yields no improvement.

**Geometric verification.** Hard DPS achieves 98.7%—exactly the probe-round upper bound—confirming that bypassing the output head analytically recovers full probe accuracy. Applying rank-16 LoRA directly to the 9 digit rows (65K params) achieves 95.2%, confirming the bottleneck is in the output head specifically.

**Generation-mode evidence.** The 9-row repair achieves 0.0% in unconstrained generation. However, four converging tests confirm the bottleneck is in output routing, not encoding: (1) probe-round in generation achieves 96.0%; (2) hard DPS achieves 72.4% (83.5% of remaining errors are format failures); (3) logit-masked generation with 9-row repair achieves 59.2%, matching constrained next-token; (4) LoRA Q/V rank-16 achieves 83.1%±7.2% in true autoregressive generation (generation gap = 0.000 across all 5 seeds).

**Mechanism of LoRA Q/V.** Direct measurements at three points in the computation graph confirm the routing-specificity interpretation. At the count-encoding layer (layer 2), LoRA Q/V leaves the probe direction unchanged (mean $|\cos| = 0.0089 \to 0.0070$). At the final layer, ridge-probe $R^2$ rises from $0.974$ to $0.998$ ($24\times$ residual amplification). Most directly, logit-lens accuracy rises from $9.3\%$ to $71.8\%$, and the correct digit's median vocabulary rank drops from 55,980 to 1 ($50{,}000\times$ improvement). This mechanism generalizes: under the same LoRA Q/V intervention, character-counting logit-lens accuracy rises from $16.2\%$ to $67.4\%$ (rank 32,265 $\to$ 16), and addition logit-lens from $24.7\%$ to $98.9\%$ (rank 21,186 $\to$ 1). The improvement magnitude tracks task difficulty: harder tasks see a larger rank reduction, confirming that LoRA Q/V corrects the same geometric bottleneck across tasks. A locus ablation — applying the same rank-16 LoRA separately to attention Q/K/V/O projections, MLPs, and combinations, then comparing logit-lens alignment — confirms Q/V yields the best logit-lens alignment of any single-projection variant; per-seed generation accuracy on entity counting alone reaches $97.0\%$, $96.5\%$, $94.5\%$ (3 seeds), showing the multi-task variance ($71.5$–$89.0\%$) is a task-mix artifact, not a reliability issue.

**Cross-model and cross-task validation.** All four models exhibit the same pattern: probes decode counts at $R^2 > 0.99$ from directions orthogonal to the output head ($|\cos| \leq 0.032$). The 9-row repair generalizes across model families where the geometric signature is present: Mistral-7B achieves 92.0% held-out. Pythia-410M reaches only 31.4%, so the repair's transferability at small scale is limited even though the orthogonality signature itself appears; we therefore scope the repair claim to mid-size and larger models. At 14B, $|\cos|$ sharpens to $0.011$ ($0.57\times$ random baseline) yet targeted repair recovers $90.3\% \pm 1.5$–scale strengthens the bottleneck but does not prevent repair.

Beyond counting, the bottleneck extends to other low-vocabulary aggregation tasks. On majority vote ("Are there more cats or dogs?"), the best probe achieves $R^2 = 0.9998$ with $|\cos| = 0.003$, and soft DPS raises accuracy from 51.4% (chance) to 100.0% (random direction: 49.4%±2.4%). On max extraction ("What is the largest number?"), baseline 15.3% improves to 98.7% with probe-round and 85.3% with 9-row repair. On multi-digit counts (10–20), baseline 5.7% improves to 93.8% with probe-round and 42.1% with fullvocab repair. These results confirm that the bottleneck operates on the *internal aggregation representation*, not on the surface task format.

**Instruct mode and natural language.** The bottleneck persists in instruct mode: first-token digit accuracy is 22% despite $R^2 \geq 0.9996$; 9-row repair recovers 99.9%. On natural-language counting (8 entity categories $\times$ 8 templates), probe-round recovers 96.3% vs. 88.7% baseline, confirming the bottleneck generalizes beyond synthetic prompts. As negative controls we evaluate two standard multi-step benchmarks: MMLU [Hendrycks et al., 2021], a multiple-choice knowledge suite, and GSM8K [Cobbe et al., 2021], grade-school math word problems. MMLU shows no bottleneck (baseline 70.2%; output-row adaptation degrades to 55.6%; $|\cos| = 0.31$–$0.48$ vs. $\leq 0.032$ for counting). On DROP [Dua et al., 2019] (single-digit subset), probe-round improves from 20.0% to 30.0% (+10 pp), suggesting partial but incomplete readout-bottleneck structure.

## 7 Discussion

**The output gap: localized and diagnosed.** Qwen3-8B encodes entity counts at $R^2 > 0.99$ but the count subspace is orthogonal to `lm_head` ($|\cos| \leq 0.032$). The strongest localization result is that updating only 9 output rows yields 93–99% held-out accuracy across Qwen3-8B and Mistral-7B, surpassing LoRA (84%, 4M params). That gradient descent independently discovers the same geometric bottleneck that DPS bypasses analytically—LoRA Q/V transforms the `lm_head` readout path: logit-lens accuracy rises from $9.3\%$ to $71.8\%$ and the correct digit's rank drops from $55{,}980$ to 1–supports the diagnosis. Our finding resonates with the superposition hypothesis [Elhage et al., 2022]: rarely-used features are stored in directions that minimize interference with frequent features, at the cost of output accessibility.

**Why LoRA Q/V, not output-head repair?** The geometric diagnosis explains the mode-specific ordering of results. The 9-row repair fixes the digit-row misalignment but not the routing dynamics: hidden states at generation steps still pass through unmodified attention and FFN layers that route away from digit tokens. LoRA Q/V corrects this upstream routing, strengthening the count signal throughout the residual stream. The locus ablation confirms Q/V yields the best alignment; the generation gap of 0.000 across all seeds confirms faithful generalization.

**Comparison with chain-of-thought.** Chain-of-thought (CoT) prompting also substantially improves entity counting, placing it alongside LoRA Q/V (83.1%±7.2%) as an effective intervention. One measurement caveat matters when comparing numbers across studies: CoT accuracy is sensitive to how the emitted reasoning is scored. Scoring the *first* integer anywhere in the output credits intermediate counting tokens rather than the model's actual answer, inflating accuracy by large margins; all generation-mode numbers in this paper score the *final* integer emitted. Mechanistically, CoT and our approach address the bottleneck through fundamentally different means: CoT externalizes the serial aggregation that the model's forward pass fails to route to the output head—each reasoning step re-encodes the count in a form the model can naturally emit. LoRA Q/V corrects the routing *internally*, enabling single-step greedy decoding without additional inference-time tokens. The two approaches are complementary: CoT requires no fine-tuning but multiplies inference cost; LoRA Q/V requires fine-tuning but adds zero inference overhead. This mechanistic distinction clarifies the paper's contribution: we are not claiming to beat CoT on accuracy, but to explain *why* CoT helps and to provide an alternative route to the same end via targeted weight modification. A full mechanistic account of CoT — e.g., whether count-to-`lm_head` alignment improves across CoT reasoning tokens, and whether filler tokens without explicit counting produce the same effect — is an open question beyond the scope of this paper.

**Limitations and scope.** (1) *Generation-mode scope:* The 9-row repair is a diagnostic instrument for constrained decoding, not a deployable fix for unconstrained generation. The deployable LoRA Q/V intervention achieves 83.1% generation accuracy but requires fine-tuning. (2) *Model scale:* Confirmed up to 14B; whether the bottleneck persists, sharpens, or resolves at frontier scales ($\geq$70B) is an open question. (3) *Architectural specificity:* Tested on three decoder-only transformer families; encoder-decoder and SSM architectures are out of scope. (4) *Entity-counting repair ceiling (60%):* Partially explained by digit-row norm competition and intra-class hidden-state diversity; discriminating these two hypotheses requires experiments varying prompt diversity at fixed count values. (5) *Task scope:* Confirmed only for low-vocabulary aggregation where the answer is a single token from a small set. Multi-step reasoning and open-ended generation are explicitly out of scope. The diagnosis *predicts* that any task where the internal representation is accurate but misaligned with the output head will exhibit the same bottleneck—confirming this on additional task families is important future work.

**Broader implications.** The finding that a specific, localized geometric misalignment explains a well-documented behavioral failure suggests a general diagnostic strategy: (1) probe for task-relevant information, (2) measure alignment with the output head, (3) target repair at the identified bottleneck. This strategy may apply to other "competence without performance" failures where models encode information they cannot express.

## 8 Conclusion

Under constrained bounded-output decoding, transformers know how to count but cannot directly emit the answer without intervention. Across three model families (0.4–14B), the residual stream encodes counts at $R^2 > 0.99$ in directions orthogonal to the output head ($|\cos| \leq 0.032$). Rewriting only 9 digit rows of `lm_head` raises next-token accuracy to 60.7–100.0% across four tasks, while LoRA Q/V rank-16 achieves 83.1% in true autoregressive generation. The bottleneck generalizes across architectures, tasks, and scales, and the geometric diagnosis makes falsifiable predictions that hold under controlled tests. These results establish counting failure as a geometric readout bottleneck: the model knows the count, but the output pathway needs realignment to express it.

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## A Full Diagnostic Verification Appendix

The main text presents the condensed diagnostic evidence. This appendix provides the full set of tables, controls, and analyses that support each claim.

### A.1 Initial Diagnostic DPS Protocol (Single-Seed)

Table 6 reports the original single-seed (seed 42) DPS experiment on Qwen3-8B entity counting that confirmed the geometric hypothesis. Under this protocol: bare count-the-X prompts, fixed single seed, and argmax over all tokens, DPS achieves 96.3%. This number differs from the mode-matched result (13.2% in Table 4) because soft DPS adds a Gaussian-shaped increment to the predicted digit's logit; however, a non-digit token (newline, space, or punctuation) wins the full-vocabulary argmax for every single baseline prompt (600/600 examples across all three seeds in entity counting). The soft boost cannot overcome a non-digit token that already leads by several logit units.

To verify the probe direction is nevertheless correct, we apply hard DPS: add $+100$ directly to the probe-predicted digit token's logit (same primary multi-seed protocol, 3 seeds $\times$ 200 prompts).

Table 6: DPS single-seed results (seed 42), $N$=300, next-token evaluation.

| Condition | Accuracy |
|---|---|
| Baseline (vanilla Qwen3-8B) | 10.3% |
| Soft DPS (layer 2, $\alpha = 5.0$) | 96.3% |
| Oracle (true count probe) | 100.0% |
| Random direction ($\alpha = 5.0$, 10 seeds) | 10.2% ± 1.9% |

Note: soft DPS success here (96.3%) vs. failure in Table 4 (13.2%) reflects protocol differences: single-seed vs. multi-seed with diverse templates.

Full single-seed DPS reproducibility (5 seeds: $94.4\%\pm1.6\%$; probe-round $95.3\%\pm1.3\%$), sensitivity analysis ($\alpha \in \{1, 5, 10, 20\}$, $\sigma \in \{0.3, 0.5, 1.0\}$), and a CoT comparison under the corrected final-integer scorer are available in the supplement.

Table 7: Evaluation protocols used in this paper.

| Protocol | Decoding mode | Argmax scope | Primary table |
|---|---|---|---|
| Probe-round | Next-token | Digit tokens 1–9 | Table 4 |
| 9-row repair | Next-token | Digit tokens 1–9 | Table 4 |
| Fullvocab repair | Next-token | Full vocabulary | Table 4 |
| DPS (hard bypass) | Next-token | Full/digit | §6 |
| Row repair (generation) | Autoregressive | Full vocabulary | Appendix A.5 |
| Probe-round (generation) | Autoregressive | Full vocabulary | Appendix A.5 |

### A.2 Evaluation Protocol Map

### A.3 Robustness Checks

- **Shuffled-label probe:** Shuffled probes ($n$=200) yield $R^2 = -0.042$ (max: 0.045) vs. $R^2 = 0.990$ for real labels ($p < 0.005$).
- **Random-direction cosine baseline:** $|\cos| = 0.013 \pm 0.011$, indistinguishable from count-probe alignment ($p = 0.79$, permutation test).
- **TOST equivalence:** The two distributions are statistically equivalent ($p < 0.001$).
- **Positive control:** Probe for the model's predicted continuation token achieves $|\cos| = 0.115$, $3.3\times$ higher than the count probe's 0.035.
- **Format robustness:** Probe $R^2$ remains above 0.98 and $|\cos|$ below 0.03 across 4 formats (raw, answer-only, assistant-prefix, full-instruct); digit accuracy ranges 9–21%.

### A.4 Depth Profile and Difficulty Breakdown

Cross-layer agreement predicts correctness: 15.5/36 layers have the right logit-lens argmax on correct prompts vs. 2.4/36 for incorrect.

### A.5 Generation-Mode Protocol Details

Under unconstrained autoregressive generation (greedy decode, max 8 tokens, final integer extracted as the model's answer):

- **Probe-round (generation):** 96.0% [94.7, 97.2] ($N$=900, 3 seeds). Probe injects count prediction at each decoding step. Upper bound diagnostic.
- **Hard DPS (generation):** 72.4% [69.6, 75.4] ($\alpha$=20). 83.5% of errors are format failures (digit not first token), not wrong-digit errors.
- **9-row repair (generation):** 0.0%. The repaired rows encode the correct answer but full-vocabulary argmax at each step is still misaligned.
- **Logit-masked generation:** 9-row repair + constrain to digit tokens: 59.2% ($N$=600, 3 seeds, range 57.5–61.5%). Matches constrained next-token accuracy, confirming the repair correctly encodes the answer.
- **LoRA Q/V rank-16 (generation):** 83.1%±7.2% (5 seeds, multi-task). Generation gap = 0.000 across all seeds.

### A.6 Necessity and Sufficiency Controls

Shuffled-digit and random-position controls confirm the specific row-token mapping is both necessary and sufficient: shuffled rows (14.0%) degrade below baseline (17.0%), and trained rows at random non-digit positions match baseline exactly.

### A.7 Logit-Gap Ceiling Model

The gap between probe-round and 9-row repair is count-magnitude-dependent (Table 8):

Table 8: Entity counting accuracy stratified by count value ($N$=600, 3 seeds).

| Count | $N$ | Probe-round | 9-row repair |
|---|---|---|---|
| 1 | 65 | 98.5% | 92.3% |
| 2 | 62 | 96.8% | 100.0% |
| 3 | 71 | 100.0% | 80.3% |
| 4 | 78 | 97.4% | 51.3% |
| 5 | 73 | 100.0% | 39.7% |
| 6 | 73 | 97.3% | 37.0% |
| 7 | 62 | 100.0% | 30.6% |
| 8 | 59 | 100.0% | 49.2% |
| 9 | 57 | 98.2% | 71.9% |

### A.8 Format Robustness Check

A 4-format robustness check (raw, answer-only, assistant-prefix, full-instruct) confirms the bottleneck is not a prompt artifact: probe $R^2$ remains above 0.98 and $|\cos|$ below 0.03 across all formats, while digit accuracy ranges 9–21%.

### A.9 Multi-Digit Extension: Counts 10–20

The bottleneck extends to multi-digit counts. On counts 10–20, baseline fullvocab is 5.7%, probe-round 93.8%, and fullvocab repair 42.1%. The geometric bottleneck operates at the token level: each digit position in a multi-digit number is independently misaligned.

### A.10 Max Extraction: A Non-Count Aggregation Task

Max-extraction ("What is the largest number in the list?") tests whether the geometric bottleneck extends beyond counting to other low-vocabulary aggregation. On Qwen3-8B: baseline 15.3%, probe-round 98.7%, 9-row repair 85.3% (3 seeds), fullvocab repair 92.7%. Hard DPS achieves 97.3%. The bottleneck is not specific to counting.

### A.11 Majority Vote: Same Bottleneck, Different Surface Task

Majority vote ("Are there more cats or dogs?") tests the bottleneck on a binary classification task. Baseline 51.4% (chance), best probe $R^2 = 0.9998$, $|\cos| = 0.003$ (lower than counting), soft DPS ($\alpha$=5) achieves 100.0%. Random direction control: 49.4%±2.4% (20 seeds). The bottleneck operates on the internal count representation regardless of output format.

### A.12 Capacity Ablation for Entity Counting

We test two candidate explanations for the 37 pp entity counting gap: conservative ridge regularization and insufficient row count. Adam fine-tuning of the same 9 digit rows achieves 67.5%±1.8%, only 10.8 pp above the ridge baseline (56.7% in the ablation replication), ruling out conservative regularization. Expanding to 59 rows (top-50 by cosine alignment plus all digit rows) yields no further improvement (67.5%±1.8%), ruling out row count as a capacity bottleneck. The remaining 31 pp gap reflects a task-level ceiling, not a fitting or capacity limitation.

### A.13 Generation-Mode Mismatch Diagnosis

The mismatch between constrained next-token accuracy and generation accuracy is further diagnosed with a routing experiment: when generation is constrained to digit tokens via logit masking, the 9-row repaired model achieves 59.2% ($N$=600, 3 seeds), exactly matching its controlled next-token accuracy and far above the unrepaired baseline (14.2% constrained). This confirms the repair correctly encodes the answer and that the unconstrained 0.0% failure is a routing artifact.

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